Lesson 61: Kernel Methods

AI/ML Mathematics

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1 Motivation and Overview

Consider the problem of classifying data points in $\mathbb{R}^2$ that are not linearly separable—for instance, points arranged in two concentric circles. A linear classifier in $\mathbb{R}^2$ cannot separate them, but if we map each point (x_1, x_2) to a higher-dimensional feature space via $\phi(x_1, x_2) = (x_1^2, x_2^2, \sqrt{2}\, x_1 x_2) \in \mathbb{R}^3$, the classes may become linearly separable. The difficulty is that such feature maps can be extremely high-dimensional: a degree-p polynomial map on $\mathbb{R}^d$ produces features in $\mathbb{R}^{\binom{d+p}{p}}$, which for $d = 1000$ and $p = 5$ exceeds 10^{13}. Computing and storing these feature vectors is infeasible.

The *kernel trick* resolves this: if the learning algorithm depends on the data only through inner products $\langle \phi(x), \phi(x') \rangle$, we can replace this inner product by a kernel function $k(x, x')$ that is computable in $O(d)$ time, bypassing the explicit feature map entirely. This lesson develops the mathematical foundations of kernel methods: the kernel trick, positive definite kernels, Mercer's theorem providing the spectral foundation, and systematic methods for constructing kernels. The tools developed in this lesson will let us build the reproducing kernel Hilbert spaces (RKHS) framework in Lesson 62 and connect kernel methods to neural network theory.

Why Kernel Methods Matter for RL/ML

- **Support vector machines:** SVMs solve a quadratic program whose objective and constraints depend on the data only through the Gram matrix $K_{ij} = k(x_i, x_j)$. Replacing k with a nonlinear kernel yields nonlinear decision boundaries without modifying the optimization algorithm.
- **Gaussian processes:** A GP prior is specified by a mean function and a kernel $k(x, x')$. The posterior predictions, uncertainty quantification, and Bayesian optimization acquisitions all flow from kernel evaluations.
- **Neural tangent kernel (NTK):** In the infinite-width limit, a neural network trained by gradient descent evolves as a linear model in a kernel regime, where the NTK $\Theta(x, x') = \langle \nabla_\theta f_\theta(x), \nabla_\theta f_\theta(x') \rangle$ governs learning dynamics.
- **Value function approximation:** Kernel-based temporal difference methods (e.g., KLSTD) represent the value function in an RKHS, enabling nonparametric approximation with convergence guarantees.

Prerequisites. Inner product spaces and the spectral theorem for compact self-adjoint operators (Lesson 7, Lesson 19). L^2 spaces and integration (Lesson 3, Lesson 5). Basic linear algebra: eigenvalues, positive semidefiniteness (Lesson 6). Hypothesis classes and empirical risk minimization (Lesson 59). Bias-variance tradeoff (Lesson 60).

Learning objectives. After completing this lesson, the student will be able to:

- *Explain* the kernel trick and *identify* when an algorithm can be kernelized.
- *Define* positive definite kernels and *prove* that inner-product kernels are positive definite.
- *Prove* closure properties of the class of positive definite kernels (sums, products, limits).
- *State* Mercer's theorem and *derive* the eigenexpansion of a continuous positive definite kernel.
- *Construct* standard kernels (polynomial, Gaussian RBF, Laplacian) and *verify* their positive definiteness.
- *Apply* kernel methods to a concrete classification or regression problem.

2 References

- Bach [2024, Chapter 4] – Kernels, positive definite functions, and Mercer's theorem in the learning theory context.
- Schölkopf and Smola [2002, Chapters 2–3] – Comprehensive treatment of kernels, feature spaces, and the kernel trick for SVMs.
- Steinwart and Christmann [2008, Chapters 5–6] – Rigorous treatment of Mercer kernels, RKHS construction, and kernel properties.
- Mohri et al. [2018, Chapter 6] – Kernel methods from the foundations of machine learning perspective.
- Mercer [1909] – Original paper on the eigenexpansion of positive definite integral kernels.

3 The Kernel Trick

The kernel trick is a computational strategy: replace explicit feature-space computations with evaluations of a kernel function. We begin by formalizing the setting.

Definition 3.1 (Feature Map)**.** Let $\mathcal{X}$ be an input space (typically a subset of $\mathbb{R}^d$) and let $\mathcal{V}$ be an inner product space (the *feature space*). A *feature map* is any function $\phi\colon \mathcal{X} \to \mathcal{V}$.

Definition 3.2 (Kernel Function)**.** Given a feature map $\phi\colon \mathcal{X} \to \mathcal{V}$, the associated *kernel function* $k\colon \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is defined by

$$k(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathcal{V}}. \tag{1}$$

The key observation is that many learning algorithms—including ridge regression, support vector machines, and principal component analysis—interact with the data $x_1, \ldots, x_n$ only through pairwise inner products $\langle \phi(x_i), \phi(x_j) \rangle$. If we can evaluate $k(x_i, x_j)$ directly, we never need to compute $\phi(x_i)$ explicitly.

Example 3.3 (Polynomial Kernel)**.** Let $\mathcal{X} = \mathbb{R}^d$ and define $k(x, x') = (1 + x^\top x')^p$ for a positive integer p. We show this is an inner-product kernel by exhibiting the feature map for $d = 2$, $p = 2$.

We have $k(x, x') = (1 + x_1 x_1' + x_2 x_2')^2$. Expanding:

$$\begin{aligned} k(x, x') &= 1 + x_1^2 (x_1')^2 + x_2^2 (x_2')^2 + 2x_1 x_1' + 2x_2 x_2' + 2x_1 x_2 x_1' x_2' \\ &= \langle \phi(x), \phi(x') \rangle, \end{aligned}$$

where $\phi(x) = (1, x_1^2, x_2^2, \sqrt{2}\, x_1, \sqrt{2}\, x_2, \sqrt{2}\, x_1 x_2) \in \mathbb{R}^6$.

More generally, for $\mathcal{X} = \mathbb{R}^d$ the feature space has dimension $\binom{d+p}{p}$, but evaluating $k(x, x')$ requires only $O(d)$ operations (one inner product plus exponentiation).

Example 3.4 (Kernelized Ridge Regression)**.** Standard ridge regression solves $\min_{w \in \mathbb{R}^D} \sum_{i=1}^n (y_i - w^\top \phi(x_i))^2 + \lambda \|w\|^2$ in the feature space $\mathbb{R}^D$, where $D = \dim(\mathcal{V})$ may be very large. Writing $\Phi = [\phi(x_1), \ldots, \phi(x_n)]^\top \in \mathbb{R}^{n \times D}$, the solution is $w^* = (\Phi^\top \Phi + \lambda I)^{-1} \Phi^\top y$, which requires $O(D^2 n)$ operations.

By the matrix inversion lemma, the prediction at a new point x is

$$f(x) = (w^*)^\top \phi(x) = k_x^\top (K + \lambda I)^{-1} y,$$

where $K_{ij} = k(x_i, x_j)$ is the $n \times n$ Gram matrix and $k_x = (k(x_1, x), \ldots, k(x_n, x))^\top$. This requires only $O(n^2 d + n^3)$ operations, independent of D.

Proposition 3.5 (Kernelization Criterion)**.** *An algorithm operating on data $x_1, \ldots, x_n \in \mathcal{X}$ via a feature map ϕ can be kernelized if and only if every computation involving the feature vectors $\phi(x_1), \ldots, \phi(x_n)$ can be expressed in terms of the entries of the Gram matrix $K_{ij} = \langle \phi(x_i), \phi(x_j) \rangle$.*

Proof. The "if" direction is immediate: if all computations reduce to entries of K, we replace each K_{ij} by $k(x_i, x_j)$.

For the "only if" direction, suppose the algorithm requires some computation that cannot be expressed through inner products alone. Then there exist two feature maps ϕ, ϕ' with $\langle \phi(x_i), \phi(x_j) \rangle = \langle \phi'(x_i), \phi'(x_j) \rangle$ for all i, j, but the algorithm produces different outputs for ϕ and ϕ'. Hence the algorithm is not determined by the kernel, and kernelization is impossible. □

The Kernel Trick in RL

In temporal difference learning with linear function approximation, the value function is $\hat{V}(s) = w^\top \phi(s)$. The TD(0) update $w \leftarrow w + \alpha \delta_t \phi(s_t)$, where $\delta_t = r_t + \gamma w^\top \phi(s_{t+1}) - w^\top \phi(s_t)$, involves ϕ only through inner products. This enables kernel TD methods that represent the value function nonparametrically as $\hat{V}(s) = \sum_{i=1}^n \alpha_i k(s_i, s)$.

The kernel trick transforms the computational challenge from working in a high-dimensional feature space to working with an $n \times n$ matrix, but it raises a fundamental question: which functions $k \colon \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ correspond to inner products in some feature space? This is answered by the theory of positive definite kernels.

4 Positive Definite Kernels

The concept of positive definiteness characterizes exactly those functions that arise as inner products in some feature space.

Definition 4.1 (Positive Definite Kernel)**.** A function $k \colon \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is *positive definite* (pd) if it is symmetric ($k(x, x') = k(x', x)$ for all $x, x' \in \mathcal{X}$) and for every $n \geq 1$, every set of points $\{x_1, \ldots, x_n\} \subseteq \mathcal{X}$, and every $\alpha \in \mathbb{R}^n$:

$$\sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k(x_i, x_j) \geq 0. \tag{2}$$

Equivalently, the *Gram matrix* $K \in \mathbb{R}^{n \times n}$ with $K_{ij} = k(x_i, x_j)$ is positive semidefinite for every finite subset of $\mathcal{X}$.

Remark 4.2. Some authors use "positive semidefinite kernel" for the condition (2) and reserve "positive definite" for strict inequality when $\alpha \neq 0$ and $x_1, \ldots, x_n$ are distinct. We follow the convention of Bach [2024] and Schölkopf and Smola [2002], using "positive definite" for the condition as stated.

Theorem 4.3 (Inner-Product Kernels are Positive Definite)**.** *If $\phi \colon \mathcal{X} \to \mathcal{V}$ is a feature map into an inner product space $\mathcal{V}$, then $k(x, x') = \langle \phi(x), \phi(x') \rangle$ is a positive definite kernel.*

Proof. Let $x_1, \ldots, x_n \in \mathcal{X}$ and $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. Then

$$\begin{aligned}
\sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j k(x_i, x_j) &= \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j \langle \phi(x_i), \phi(x_j) \rangle \\
&= \left\langle \sum_{i=1}^n \alpha_i \phi(x_i), \sum_{j=1}^n \alpha_j \phi(x_j) \right\rangle \\
&= \left\| \sum_{i=1}^n \alpha_i \phi(x_i) \right\|^2 \geq 0,
\end{aligned}$$

where the second equality uses bilinearity of the inner product and the last step uses positive definiteness of the norm. □

The converse also holds: every positive definite kernel arises from an inner product in some feature space. This is the content of the RKHS construction in Lesson 62 (Aronszajn's theorem).

Characterization of Positive Definite Kernels

A symmetric function $k\colon \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is positive definite if and only if there exists a Hilbert space $\mathcal{V}$ and a feature map $\phi\colon \mathcal{X} \to \mathcal{V}$ such that $k(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathcal{V}}$. The "if" direction is Theorem 4.3; the "only if" direction is Aronszajn's theorem (Lesson 62, Theorem 3.5).

We now establish that the class of pd kernels is closed under several useful operations.

Theorem 4.4 (Closure Properties of PD Kernels)**.** *Let k_1, k_2 be positive definite kernels on $\mathcal{X} \times \mathcal{X}$. Then the following are also positive definite:*

(i) ***Sum:*** $k(x, x') = k_1(x, x') + k_2(x, x')$.

(ii) ***Scalar multiple:*** $k(x, x') = c\, k_1(x, x')$ *for any* $c \geq 0$.

(iii) ***Product:*** $k(x, x') = k_1(x, x') \cdot k_2(x, x')$.

(iv) ***Pointwise limit:*** *If k_m is pd for each m and $k_m(x, x') \to k(x, x')$ for all x, x', then k is pd.*

Proof. Let $x_1, \ldots, x_n \in \mathcal{X}$ and $\alpha \in \mathbb{R}^n$. Let $K^{(1)}, K^{(2)}$ denote the Gram matrices of k_1, k_2.

(i) Sum. The Gram matrix is $K = K^{(1)} + K^{(2)}$. Then $\alpha^\top K \alpha = \alpha^\top K^{(1)} \alpha + \alpha^\top K^{(2)} \alpha \geq 0 + 0 = 0$.

(ii) Scalar multiple. The Gram matrix is $cK^{(1)}$. Then $\alpha^\top (cK^{(1)}) \alpha = c\, \alpha^\top K^{(1)} \alpha \geq 0$ since $c \geq 0$.

(iii) Product (Schur product theorem). The Gram matrix is the Hadamard (entrywise) product $K = K^{(1)} \circ K^{(2)}$. By the spectral theorem, write $K^{(2)} = \sum_{l=1}^n \mu_l v_l v_l^\top$ where $\mu_l \geq 0$ (since $K^{(2)} \succeq 0$). Then

$$K_{ij} = K_{ij}^{(1)} K_{ij}^{(2)} = K_{ij}^{(1)} \sum_{l=1}^n \mu_l (v_l)_i (v_l)_j = \sum_{l=1}^n \mu_l (v_l)_i (v_l)_j K_{ij}^{(1)}.$$

Hence $\alpha^\top K \alpha = \sum_{l=1}^n \mu_l \sum_{i,j} \alpha_i (v_l)_i K_{ij}^{(1)} \alpha_j (v_l)_j = \sum_{l=1}^n \mu_l\, \beta_l^\top K^{(1)} \beta_l$, where $\beta_l = (\alpha_1 (v_l)_1, \ldots, \alpha_n (v_l)_n)$. Since $K^{(1)} \succeq 0$ and $\mu_l \geq 0$, each term is nonnegative, so $\alpha^\top K \alpha \geq 0$.

(iv) Pointwise limit. The Gram matrix $K^{(m)}$ of k_m satisfies $\alpha^\top K^{(m)} \alpha \geq 0$ for each m. Since $K_{ij}^{(m)} \to K_{ij}$ entrywise as $m \to \infty$, we obtain $\alpha^\top K \alpha = \lim_{m \to \infty} \alpha^\top K^{(m)} \alpha \geq 0$. □

Example 4.5 (Linear Kernel)**.** The *linear kernel* on $\mathbb{R}^d$ is $k(x, x') = x^\top x'$. This corresponds to the identity feature map $\phi(x) = x$. It is pd by Theorem 4.3.

Example 4.6 (Polynomial Kernel – Positive Definiteness)**.** The polynomial kernel $k(x, x') = (1 + x^\top x')^p$ for $p \in \mathbb{N}$ is pd. To see this, write $k(x, x') = \sum_{j=0}^p \binom{p}{j} (x^\top x')^j$. The linear kernel $k_0(x, x') = x^\top x'$ is pd by the previous example. By closure under products (iii), $(x^\top x')^j$ is pd for each $j \geq 1$. The constant function $k(x, x') = 1$ is pd (Gram matrix is all-ones, which is psd). By closure under sums (i) and scalar multiples (ii), k is pd.

Example 4.7 (Gaussian RBF Kernel)**.** The *Gaussian radial basis function* (RBF) kernel on $\mathbb{R}^d$ is

$$k(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right), \tag{3}$$

where $\sigma > 0$ is the *bandwidth* parameter. This is perhaps the most widely used kernel in practice. We prove its positive definiteness in Section 6 (Theorem 6.3).

Not Every Symmetric Function is a Kernel

The function $k(x, x') = -\|x - x'\|^2$ is symmetric but *not* positive definite. For $n = 2$ points x_1, x_2 with $\alpha = (1, -1)^\top$: $\alpha^\top K\alpha = -2\|x_1 - x_2\|^2 < 0$ whenever $x_1 \neq x_2$. One must always verify positive definiteness.

The closure properties allow us to build a rich algebra of kernels from simple building blocks. We next develop the spectral theory of positive definite kernels, which reveals their deep structure.

5 Mercer's Theorem

Mercer's theorem provides a spectral decomposition for continuous positive definite kernels, analogous to the eigendecomposition of positive semidefinite matrices. This decomposition is the foundation of the canonical feature map and underpins the RKHS construction of Lesson 62.

Assumption 5.1 (Setting for Mercer's Theorem)**.** Let $\mathcal{X}$ be a compact subset of $\mathbb{R}^d$ and let μ be a finite Borel measure on $\mathcal{X}$ with $\operatorname{supp}(\mu) = \mathcal{X}$. Let $k\colon \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be continuous.

Under these assumptions, the kernel defines an integral operator.

Definition 5.2 (Integral Operator)**.** The *integral operator* $T_k\colon L^2(\mathcal{X}, \mu) \to L^2(\mathcal{X}, \mu)$ associated with kernel k is

$$(T_k f)(x) = \int_{\mathcal{X}} k(x, x') f(x')\, d\mu(x'). \tag{4}$$

Lemma 5.3 (Properties of T_k)**.** *Under Assumption 5.1, if k is continuous and positive definite, then T_k is:*

(i) a bounded linear operator on $L^2(\mathcal{X}, \mu)$;

(ii) self-adjoint: $\langle T_k f, g\rangle_{L^2} = \langle f, T_k g\rangle_{L^2}$;

(iii) positive: $\langle T_k f, f\rangle_{L^2} \geq 0$ for all f;

(iv) compact.

Proof. **(i)** Since k is continuous on $\mathcal{X} \times \mathcal{X}$ and $\mathcal{X}$ is compact, $M := \sup_{x,x'} |k(x, x')| < \infty$. By the Cauchy–Schwarz inequality,

$$|(T_k f)(x)| \leq \int_{\mathcal{X}} |k(x, x')||f(x')|\, d\mu(x') \leq M\|f\|_{L^1} \leq M\mu(\mathcal{X})^{1/2}\|f\|_{L^2},$$

so $\|T_k f\|_{L^2} \leq M\mu(\mathcal{X})\|f\|_{L^2}$, establishing boundedness.

(ii) By symmetry of k and Fubini's theorem:

$$\begin{aligned}\langle T_k f, g\rangle_{L^2} &= \int_{\mathcal{X}} \int_{\mathcal{X}} k(x, x') f(x') g(x)\, d\mu(x')\, d\mu(x) \\ &= \int_{\mathcal{X}} f(x') \int_{\mathcal{X}} k(x', x) g(x)\, d\mu(x)\, d\mu(x') = \langle f, T_k g\rangle_{L^2}.\end{aligned}$$

(iii) For positive definiteness, we approximate. For $f \in L^2(\mathcal{X}, \mu)$:

$$\langle T_k f, f \rangle_{L^2} = \int_{\mathcal{X}} \int_{\mathcal{X}} k(x, x') f(x) f(x') \, d\mu(x) \, d\mu(x').$$

This double integral is the continuous analogue of the quadratic form $\alpha^\top K \alpha$. To verify nonnegativity, approximate f by simple functions $f_n = \sum_{i=1}^{m_n} c_i \mathbf{1}_{A_i}$. For simple functions, the integral reduces to $\sum_{i,j} c_i c_j \int_{A_i} \int_{A_j} k(x, x') \, d\mu(x) \, d\mu(x')$. By a discretization argument (partitioning each A_i into small sets and applying the pd condition on point evaluations), each such sum is nonnegative. Taking limits (using dominated convergence), $\langle T_k f, f \rangle_{L^2} \geq 0$.

(iv) Since k is continuous on the compact set $\mathcal{X} \times \mathcal{X}$, it is uniformly continuous. The family $\{x' \mapsto k(x, x') : x \in \mathcal{X}\}$ is equicontinuous and uniformly bounded in L^2. Therefore T_k maps the unit ball to an equicontinuous family, which is precompact in L^2 by the Arzela–Ascoli theorem. Hence T_k is compact. □

Theorem 5.4 (Mercer's Theorem)**.** *Under Assumption 5.1, if k is continuous and positive definite, then there exists a sequence of eigenfunctions $\{\phi_j\}_{j=1}^\infty$ forming an orthonormal basis of $\overline{\text{Range}(T_k)}$ in $L^2(\mathcal{X}, \mu)$, with corresponding eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$, such that:*

$$k(x, x') = \sum_{j=1}^{\infty} \lambda_j \, \phi_j(x) \, \phi_j(x'), \tag{5}$$

where the convergence is absolute and uniform on $\mathcal{X} \times \mathcal{X}$. Moreover, $T_k \phi_j = \lambda_j \phi_j$ and $\sum_{j=1}^\infty \lambda_j = \int_{\mathcal{X}} k(x, x) \, d\mu(x) < \infty$.

Proof. By Lemma 5.3, T_k is a compact, self-adjoint, positive operator on the separable Hilbert space $L^2(\mathcal{X}, \mu)$.

Step 1: Spectral decomposition. By the spectral theorem for compact self-adjoint operators (Lesson 19), there exists an orthonormal system $\{\phi_j\}_{j \geq 1}$ of eigenfunctions with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$ satisfying $\lambda_j \to 0$, such that

$$T_k f = \sum_{j=1}^{\infty} \lambda_j \langle f, \phi_j \rangle_{L^2} \phi_j$$

for all $f \in L^2(\mathcal{X}, \mu)$. Since T_k is positive, all $\lambda_j \geq 0$.

Step 2: L^2 expansion of the kernel. Fix $x \in \mathcal{X}$ and consider $k_x := k(x, \cdot) \in L^2(\mathcal{X}, \mu)$ (this holds since k is bounded on the compact set $\mathcal{X} \times \mathcal{X}$). Then

$$(T_k \phi_j)(x) = \int_{\mathcal{X}} k(x, x') \phi_j(x') \, d\mu(x') = \lambda_j \phi_j(x).$$

Also, $\langle k_x, \phi_j \rangle_{L^2} = \int_{\mathcal{X}} k(x, x') \phi_j(x') \, d\mu(x') = \lambda_j \phi_j(x)$. Hence the Fourier expansion of k_x in L^2 gives

$$k(x, \cdot) = \sum_{j=1}^{\infty} \langle k_x, \phi_j \rangle_{L^2} \phi_j(\cdot) = \sum_{j=1}^{\infty} \lambda_j \phi_j(x) \phi_j(\cdot),$$

with convergence in $L^2(\mathcal{X}, \mu)$. Evaluating at x' (which requires pointwise convergence, established below) gives (5).

Step 3: Uniform convergence. Define the partial sums $k_N(x, x') = \sum_{j=1}^N \lambda_j \phi_j(x) \phi_j(x')$. Since each ϕ_j is continuous (as $\phi_j = \lambda_j^{-1} T_k \phi_j$ and T_k maps L^2 functions to continuous functions when k is continuous), k_N is continuous.

For the diagonal, by Parseval's identity applied to k_x:

$$\sum_{j=1}^{\infty} \lambda_j^2 \phi_j(x)^2 = \sum_{j=1}^{\infty} |\langle k_x, \phi_j \rangle_{L^2}|^2 \leq \|k_x\|_{L^2}^2 < \infty.$$

Moreover, $k(x,x) - k_N(x,x) = \sum_{j>N} \lambda_j \phi_j(x)^2 \geq 0$ (since $\lambda_j \geq 0$). By Dini's theorem, since $k(x,x) - k_N(x,x)$ is a decreasing sequence of continuous functions converging pointwise to 0 on the compact set $\mathcal{X}$, the convergence is uniform on the diagonal.

For the off-diagonal, by the Cauchy–Schwarz inequality:

$$\begin{aligned} |k(x,x') - k_N(x,x')| &= \left| \sum_{j>N} \lambda_j \phi_j(x) \phi_j(x') \right| \\ &\leq \left(\sum_{j>N} \lambda_j \phi_j(x)^2 \right)^{1/2} \left(\sum_{j>N} \lambda_j \phi_j(x')^2 \right)^{1/2} \\ &= \sqrt{(k(x,x) - k_N(x,x))(k(x',x') - k_N(x',x'))}. \end{aligned}$$

Since the diagonal convergence is uniform, the right side converges uniformly to 0 on $\mathcal{X} \times \mathcal{X}$.

Step 4: Trace formula. Integrating the diagonal: $\sum_{j=1}^{\infty} \lambda_j = \sum_{j=1}^{\infty} \lambda_j \int_{\mathcal{X}} \phi_j(x)^2 \, d\mu(x) = \int_{\mathcal{X}} k(x,x) \, d\mu(x) < \infty$, using uniform convergence to exchange sum and integral. □

Mercer's Canonical Feature Map

Mercer's theorem provides a *canonical feature map*. Define $\Phi \colon \mathcal{X} \to \ell^2$ by

$$\Phi(x) = \left(\sqrt{\lambda_1} \, \phi_1(x), \ \sqrt{\lambda_2} \, \phi_2(x), \ \ldots \right).$$

Then $\langle \Phi(x), \Phi(x') \rangle_{\ell^2} = \sum_{j=1}^{\infty} \lambda_j \phi_j(x) \phi_j(x') = k(x,x')$. This is the unique (up to unitary equivalence) feature map into a separable Hilbert space.

Example 5.5 (Eigenexpansion of the Gaussian Kernel on $[0,1]$)**.** Consider the Gaussian kernel $k(x,x') = e^{-(x-x')^2/(2\sigma^2)}$ on $\mathcal{X} = [0,1]$ with Lebesgue measure. While no closed-form eigenfunctions exist in general, the eigenvalues λ_j decay exponentially fast. Specifically, for the Gaussian kernel on $\mathbb{R}$ (with Gaussian measure), the eigenfunctions are Hermite functions and the eigenvalues satisfy $\lambda_j = O(c^j)$ for some $0 < c < 1$ depending on σ.

For numerical illustration with $\sigma = 0.3$, the first few normalized eigenvalues are approximately $\lambda_1 \approx 0.42$, $\lambda_2 \approx 0.28$, $\lambda_3 \approx 0.15$, $\lambda_4 \approx 0.07$, with rapid decay thereafter. The eigenfunctions ϕ_j oscillate with increasing frequency, analogous to Fourier modes. The rapid eigenvalue decay means that a low-rank approximation $k(x,x') \approx \sum_{j=1}^{J} \lambda_j \phi_j(x) \phi_j(x')$ is accurate even for moderate J.

Mercer's Theorem and the NTK

The neural tangent kernel $\Theta(x,x') = \langle \nabla_\theta f_\theta(x), \nabla_\theta f_\theta(x') \rangle$ is positive definite (by Theorem 4.3 with the gradient as feature map). In the infinite-width limit, the NTK is deterministic and continuous, so Mercer's theorem applies. The eigenvalues of the NTK integral operator determine the learning rates of different modes: the network learns eigenfunctions with large λ_j first (spectral bias), explaining why neural networks tend to learn low-frequency patterns before high-frequency ones.

Mercer's theorem tells us that every continuous pd kernel has a spectral decomposition. We now turn to the practical question of constructing kernels for specific applications.

6 Kernel Construction

We develop systematic methods for building positive definite kernels, starting from basic building blocks and using the closure properties of Theorem 4.4.

6.1 Kernels from Feature Maps

The most direct construction: choose a feature map $\phi\colon \mathcal{X} \to \mathbb{R}^D$ and define $k(x, x') = \phi(x)^\top \phi(x')$.

Example 6.1 (Bag-of-Words Kernel)**.** For text classification, let $\mathcal{X}$ be a set of documents and define $\phi(x) \in \mathbb{R}^V$ as the word-count vector (where V is the vocabulary size). Then $k(x, x') = \phi(x)^\top \phi(x')$ counts the total number of shared word occurrences. This is pd by Theorem 4.3.

6.2 Polynomial Kernels

We have already shown that $k(x, x') = (1 + x^\top x')^p$ is pd (Example 4.6). More generally:

Proposition 6.2 (General Polynomial Kernel)**.** *For $c \geq 0$ and $p \in \mathbb{N}$, the kernel $k(x, x') = (c + x^\top x')^p$ on $\mathbb{R}^d$ is positive definite.*

Proof. The constant kernel $k_c(x, x') = c$ is pd (its Gram matrix cJ_n, where J_n is the all-ones matrix, is psd since $J_n = \mathbf{1}\mathbf{1}^\top$). The linear kernel $k_0(x, x') = x^\top x'$ is pd. By closure under sums, $k_c + k_0$ is pd. By closure under products, $(c + x^\top x')^p$ is pd. □

6.3 The Gaussian RBF Kernel

Theorem 6.3 (Positive Definiteness of the Gaussian Kernel)**.** *The Gaussian RBF kernel $k(x, x') = \exp(-\|x - x'\|^2/(2\sigma^2))$ is positive definite on $\mathbb{R}^d$ for any $\sigma > 0$.*

Proof. We proceed in three steps.

Step 1. Expand the squared distance:

$$k(x, x') = \exp\left(-\frac{\|x\|^2}{2\sigma^2}\right) \exp\left(\frac{x^\top x'}{\sigma^2}\right) \exp\left(-\frac{\|x'\|^2}{2\sigma^2}\right).$$

Step 2. Show that $k_0(x, x') = \exp(x^\top x'/\sigma^2)$ is pd. Write the Taylor expansion:

$$\exp\left(\frac{x^\top x'}{\sigma^2}\right) = \sum_{p=0}^{\infty} \frac{1}{p!} \left(\frac{x^\top x'}{\sigma^2}\right)^p = \sum_{p=0}^{\infty} \frac{1}{p!\,\sigma^{2p}} (x^\top x')^p.$$

Each term $(x^\top x')^p$ is pd (as the p-fold product of the linear kernel). Each coefficient $1/(p!\,\sigma^{2p}) > 0$. Finite sums of pd kernels with nonnegative coefficients are pd by Theorem 4.4(i)–(ii). The infinite sum converges pointwise, so by Theorem 4.4(iv), k_0 is pd.

Step 3. Define $g(x) = \exp(-\|x\|^2/(2\sigma^2))$. Then $k(x, x') = g(x)\, k_0(x, x')\, g(x')$. For any $\alpha \in \mathbb{R}^n$:

$$\sum_{i,j} \alpha_i \alpha_j k(x_i, x_j) = \sum_{i,j} (\alpha_i g(x_i))(\alpha_j g(x_j)) k_0(x_i, x_j) = \sum_{i,j} \beta_i \beta_j k_0(x_i, x_j) \geq 0,$$

where $\beta_i = \alpha_i g(x_i)$, and the last inequality holds because k_0 is pd. Hence k is pd. □

Remark 6.4 (Infinite-Dimensional Feature Space). The Mercer eigenvalues of the Gaussian kernel are all strictly positive (the kernel is *strictly* positive definite on any finite point set with distinct points). This means the feature space is infinite-dimensional—no finite-dimensional feature map can reproduce the Gaussian kernel exactly. The power of the kernel trick is that we can work with this infinite-dimensional feature space implicitly through $O(1)$ kernel evaluations.

6.4 Laplacian Kernel

Proposition 6.5 (Laplacian Kernel)**.** *The Laplacian kernel* $k(x, x') = \exp(-\|x - x'\|_1/\sigma)$ *on* $\mathbb{R}^d$ *is positive definite for* $\sigma > 0$.

Proof. Since $\|x - x'\|_1 = \sum_{j=1}^d |x_j - x'_j|$, we have $k(x, x') = \prod_{j=1}^d \exp(-|x_j - x'_j|/\sigma)$. It suffices to show that $k_1(s, t) = \exp(-|s - t|/\sigma)$ on $\mathbb{R}$ is pd. This follows from the Fourier representation: $e^{-|u|/\sigma} = \frac{2/\sigma}{\pi} \int_0^\infty \frac{\cos(\omega u)}{1/\sigma^2 + \omega^2}\, d\omega$ for $u \in \mathbb{R}$. Since each $\cos(\omega(s - t))$ defines a pd kernel (being $\langle(\cos \omega s, \sin \omega s), (\cos \omega t, \sin \omega t)\rangle$) and we integrate with a nonnegative weight, k_1 is pd. By closure under products, k is pd. □

6.5 Operations Preserving Positive Definiteness

Beyond sums, products, and limits, several other operations preserve positive definiteness.

Proposition 6.6 (Additional Closure Operations)**.** *Let* k *be a pd kernel on* $\mathcal{X}$.

(i) ***Composition with a power series:*** *If* $f(t) = \sum_{p=0}^\infty a_p t^p$ *with* $a_p \geq 0$ *for all* p *and the series converges for* $t \geq 0$*, then* $f \circ k$ *is pd.*

(ii) ***Restriction:*** *If* $\mathcal{Y} \subseteq \mathcal{X}$*, then* $k|_{\mathcal{Y}\times\mathcal{Y}}$ *is pd.*

(iii) ***Direct sum:*** *If* k_1 *is pd on* $\mathcal{X}_1$ *and* k_2 *is pd on* $\mathcal{X}_2$*, then* $k((x_1, x_2), (x'_1, x'_2)) = k_1(x_1, x'_1) + k_2(x_2, x'_2)$ *is pd on* $\mathcal{X}_1 \times \mathcal{X}_2$.

(iv) ***Tensor product:*** $k((x_1, x_2), (x'_1, x'_2)) = k_1(x_1, x'_1) \cdot k_2(x_2, x'_2)$ *is pd on* $\mathcal{X}_1 \times \mathcal{X}_2$.

Proof. **(i)** $f(k(x, x')) = \sum_{p=0}^\infty a_p k(x, x')^p$. Each k^p is pd (closure under products), $a_p \geq 0$ (closure under scalar multiples), partial sums are pd (closure under sums), and the pointwise limit is pd (closure under limits).

(ii) The Gram matrix of $k|_{\mathcal{Y}}$ on points $y_1, \ldots, y_n \in \mathcal{Y} \subseteq \mathcal{X}$ is the same as the Gram matrix of k on these points, which is psd.

(iii)–(iv) For direct sums: the Gram matrix is $K_1 + K_2$ where $K_1, K_2 \succeq 0$. For tensor products: the Gram matrix is $K_1 \circ K_2$ (Hadamard product), which is psd by the Schur product theorem. □

Example 6.7 (Constructing Kernels by Composition)**.** Starting from the linear kernel $k_0(x, x') = x^\top x'$:

- $\exp(k_0) = \exp(x^\top x')$ is pd by (i) with $f = \exp$.
- $(1 + k_0)^p$ is pd by (i) with $f(t) = (1 + t)^p = \sum_{j=0}^p \binom{p}{j} t^j$.
- $\frac{1}{1-ck_0}$ for $0 < c < 1/\sup_x \|x\|^2$ is pd by (i) with $f(t) = 1/(1 - ct) = \sum_{p=0}^\infty c^p t^p$.

Kernel Selection Matters

The choice of kernel encodes prior assumptions about the target function's smoothness. The Gaussian kernel with small bandwidth σ produces very flexible models (low bias, high variance), while large σ yields smoother models (high bias, low variance). The Laplacian kernel produces rougher functions than the Gaussian kernel (functions in the Laplacian RKHS are only once differentiable, while the Gaussian RKHS contains C^∞ functions). Kernel selection should be guided by domain knowledge about the target function's regularity.

With a rich toolkit for constructing kernels, we are ready to develop the Hilbert space of functions associated with each kernel. This is the subject of Lesson 62.

7 Exercises

Exercise 7.1 (Positive Definiteness Verification). $(*)$
Let $k(x, x') = \min(x, x')$ for $x, x' \in [0, \infty)$. Show that k is positive definite by exhibiting an explicit feature map $\phi\colon [0, \infty) \to L^2([0, \infty))$ such that $k(x, x') = \langle \phi(x), \phi(x') \rangle_{L^2}$.

Hint: Consider $\phi(x) = \mathbf{1}_{[0,x]}$.

Exercise 7.2 (Gram Matrix Computation). $(*)$
For the data points $x_1 = (1, 0)^\top$, $x_2 = (0, 1)^\top$, $x_3 = (1, 1)^\top$ in $\mathbb{R}^2$, compute the 3×3 Gram matrix K for:

(a) the polynomial kernel $k(x, x') = (1 + x^\top x')^2$;

(b) the Gaussian RBF kernel $k(x, x') = \exp(-\|x - x'\|^2/2)$.

Verify that both matrices are positive semidefinite by computing their eigenvalues.

Exercise 7.3 (Closure Under Exponentiation). $(**)$
Let k be a positive definite kernel on $\mathcal{X}$. Prove that for any $c > 0$, the kernel $k_c(x, x') = \exp(c\, k(x, x'))$ is positive definite. Use this to give an alternative proof that the Gaussian kernel is positive definite.

Hint: First show $\exp(c\, x^\top x')$ is pd using the Taylor series, then relate the Gaussian kernel to this.

Exercise 7.4 (Non-Example: The Distance Kernel). $(**)$
Show that $k(x, x') = \|x - x'\|$ on $\mathbb{R}^d$ is *not* positive definite for $d \geq 1$. Find the smallest n and a specific point configuration $\{x_1, \ldots, x_n\} \subset \mathbb{R}$ for which the Gram matrix has a negative eigenvalue.

Exercise 7.5 (Kernel PCA Feature Extraction). $(**)$
Given data $x_1, \ldots, x_n \in \mathbb{R}^d$ and a pd kernel k, the kernel PCA algorithm extracts features from the eigendecomposition of the centered Gram matrix $\tilde{K} = HKH$ where $H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$ is the centering matrix.

(a) Show that $\tilde{K}$ is positive semidefinite.

(b) If $\tilde{K} = U \Lambda U^\top$ is the eigendecomposition with $\Lambda = \operatorname{diag}(\mu_1, \ldots, \mu_n)$ and $\mu_1 \geq \cdots \geq \mu_n \geq 0$, show that the j-th kernel PCA coordinate of x_i is $\sqrt{\mu_j}\, U_{ij}$.

(c) Explain why kernel PCA with a Gaussian kernel can separate data that linear PCA cannot.

Exercise 7.6 (Random Features Approximation). $(***)$
Rahimi and Recht [2008] showed that for shift-invariant kernels $k(x, x') = h(x - x')$ on $\mathbb{R}^d$, one can approximate the kernel using random features. By Bochner's theorem, h is the Fourier transform of a nonnegative measure Λ: $h(\delta) = \int_{\mathbb{R}^d} e^{i\omega^\top \delta}\, d\Lambda(\omega)$.

(a) Show that if $\omega \sim \Lambda/\Lambda(\mathbb{R}^d)$, then $k(x, x') = \Lambda(\mathbb{R}^d) \cdot \mathbb{E}[\cos(\omega^\top x)\cos(\omega^\top x') + \sin(\omega^\top x)\sin(\omega^\top x')]$.

(b) Define the random feature map $\hat{\phi}(x) = \sqrt{2/D}\,(\cos(\omega_1^\top x), \ldots, \cos(\omega_D^\top x))$ where $\omega_1, \ldots, \omega_D \sim \Lambda$ i.i.d. Show that $\mathbb{E}[\hat{\phi}(x)^\top \hat{\phi}(x')] = k(x, x')$.

(c) Discuss the tradeoff between the number of random features D and the approximation quality. How does this relate to the kernel trick vs. explicit feature maps?

8 Summary

- The **kernel trick** replaces inner products in a (possibly infinite-dimensional) feature space with evaluations of a kernel function $k(x, x')$, reducing computation from $O(D)$ in feature dimension to $O(d)$ in input dimension.
- A **positive definite kernel** is a symmetric function whose Gram matrix is positive semidefinite for every finite point set. This is precisely the class of functions that arise as inner products in some feature space.
- The class of pd kernels is **closed under sums, products, scalar multiples, and pointwise limits** (Theorem 4.4), enabling systematic kernel construction from simple building blocks.
- **Mercer's theorem** provides a spectral decomposition $k(x, x') = \sum_j \lambda_j \phi_j(x)\phi_j(x')$ for continuous pd kernels on compact domains, with uniform convergence and nonnegative eigenvalues.
- The **canonical feature map** $\Phi(x) = (\sqrt{\lambda_j}\phi_j(x))_{j\geq 1}$ maps into ℓ^2 and recovers the kernel as an inner product: $k(x, x') = \langle \Phi(x), \Phi(x') \rangle_{\ell^2}$.
- The **Gaussian RBF kernel** $\exp(-\|x - x'\|^2/(2\sigma^2))$ is positive definite (proved via Taylor expansion of exp), has an infinite-dimensional feature space, and its eigenvalues decay exponentially.
- **Kernel selection** encodes smoothness assumptions: the Gaussian kernel produces C^∞ functions, the Laplacian kernel produces rougher functions, and polynomial kernels constrain the function class to a finite-dimensional space.

9 Further Reading

- **Lesson 62** develops the reproducing kernel Hilbert space (RKHS) associated with each pd kernel, establishes the representer theorem, and applies it to kernel ridge regression. The RKHS provides the function space in which kernel methods operate.
- Schölkopf and Smola [2002] provide a comprehensive treatment of kernel methods for machine learning, including support vector machines, kernel PCA, and kernel Fisher discriminant analysis.
- Steinwart and Christmann [2008] give a rigorous mathematical treatment of kernels and support vector machines, including universal kernels and approximation properties.
- Rahimi and Recht [2008] introduce random features as a scalable approximation to kernel methods, bridging the gap between explicit feature maps and the kernel trick.
- Jacot et al. [2018] establish the neural tangent kernel framework, connecting wide neural networks to kernel methods and enabling the analysis of neural network training via kernel theory.
- The original paper by Mercer [1909] established the eigenexpansion of positive definite integral kernels in the context of integral equations, decades before the machine learning applications. Aronszajn [1950] later developed the abstract RKHS theory that unified and generalized Mercer's work.

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